

Shobitha A Rao

Genetics Postgraduate • Molecular Biology • Genomics • Bioinformatics

CONTACT

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SKILLS

- Molecular Biology**
- DNA & RNA Isolation
 - PCR
 - Agarose Gel Electrophoresis
 - Gel Prep & Doc
 - Primer Handling
 - NanoDrop
- Protein Analysis**
- SDS-PAGE
 - Western Blotting
 - Protein Extraction & Quantification
- Bioinformatics**
- FastQC, fastp, MultiQC
 - BLAST, QUAST, BUSCO
 - MaSuRCA, SPAdes, SOAPdenovo2
 - Molecular Docking
 - Augustus
 - Linux/Unix
- Cell & Microbiology**
- Cell Culture
 - Streak Plating
 - Microbial Culturing
 - Media Preparation
 - Aseptic Techniques
 - Gram Staining
- Lab Techniques**
- Centrifugation
 - Micropipetting
 - Buffer/Reagent Prep
 - GLP
- Software Skills**
- MS Office
 - Google Workspace
 - Notebook LM

LANGUAGES

English, Hindi, Kannada (Fluent)

SOFT SKILLS

- Teamwork
- Presentation
- Communication
- Time Management
- Leadership
- Problem Solving

PROFILE

Genetics postgraduate with a keen interest in Molecular Biology, Genomics, and Bio-informatics. Skilled in laboratory techniques, Genetic analysis, Data interpretation, and Scientific research. Looking for opportunities to apply my knowledge and contribute to research that involves molecular mechanisms, genomics, and biotechnological innovation.

EDUCATION

- Masters in Genetics** 2024 – 2026
SRM Institute of Science and Technology, Kattankulathur
CGPA: 7.5
- Bachelors in Biotechnology** 2020 – 2024
NMAMIT, University in Nitte, Karnataka
CGPA: 6.8

ACADEMIC PROJECTS

- Whole Genome Sequencing and De Novo Assembly of *Caesalpinia bonducella* L.**
- Genomic DNA Isolation & Quality Check:** Isolated high-molecular-weight genomic DNA from leaf tissue using the modified CTAB method and confirmed purity and structural integrity via 1% agarose gel electrophoresis.
 - Read Preprocessing & Filtering:** Evaluated raw sequencing data using FastQC, removed adapter sequences and low-quality bases via Cutadapt and Sickle, and filtered organellar reads using STAR to prepare cleaned nuclear datasets.
 - De Novo & Organellar Assembly:** Executed de novo nuclear assemblies using MaSuRCA and SPAdes on Linux HPC, alongside reconstructing circular chloroplast and mitochondrial scaffolds with GetOrganelle.
 - Completeness Assessment & Annotation:** Assessed gene space completeness with BUSCO, polished draft assemblies via Pilon, and predicted protein-coding genes using AUGUSTUS and NCBI BLAST to map secondary metabolite biosynthetic enzymes.
- Study on Efficacy of Hydrogels in Soil Water Retention and Plant Growth**
- Synthesized biodegradable hydrogel beads using sodium alginate with natural gums (guar gum, locust bean gum, bacterial cellulose), achieving a water-holding capacity of 82% and re-swelling capacity of 83.33%.
 - Demonstrated that SLGC hydrogels improved germination and sustained growth in radish, green gram, paddy, and spinach under reduced watering conditions.
 - Conducted systematic soil water retention studies including swelling ratio (72%), water retention (78.95%), and pH stability analysis for hydrogel performance evaluation.
 - Validated hydrogels as sustainable soil conditioners through tray and pot studies under water-deficit conditions.

LABORATORY EXPERIENCE

- Assisted in sample preparations, data collection, and quality control experiments.
- Operated and calibrated lab instruments to ensure precision.
- Conducted literature reviews supporting ongoing research publications.
- Followed SOPs, documented results, and maintained laboratory logs accurately.

CERTIFICATIONS & ACTIVITIES

- International Conference on Bioengineering, SRM – Student Support Team Member
- Genetic Engineering Research Symposium, SRM – Poster Presentation
- Internship at NUCSER – July 2023

PUBLICATIONS

Biopolymeric Hydrogel as an Eco-Friendly Soil Amendment for Water Conservation and Sustainable Agricultural Applications May 2026
International Journal of Aquatic Research and Environmental Studies, 6(S1), 493–516
Authors: D. S, C. S Tantry, D. M. Chandy, E. R. Nyirenda, S. Shetty, Z. Shaikh, Shobitha. A. Rao, M. B. Krishnaswamy, M. Kumar G T, V. S. Muddappa
[View Publication](#)